Kalman Filter & Bayes Filter: IMU Yaw Estimation and Vehicle State Classification

Viviane Solomon

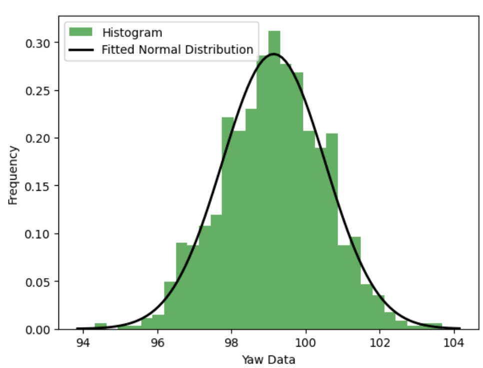
Abstract— **This lab utilizes yaw angle data acquired from a BNO055 IMU and NuScenes open-source autonomous vehicle data. The IMU data was obtained from a mobile robot, and the datasets of [x y theta] coordinates were collected at different temporal resolutions (1/13 second for IMU data and 0.5 second for NuScenes data). It should be noted that the IMU yaw angle is referenced to the East (0 degrees) and North (270 degrees).**

I. Procedure

A. OneD Kalman Filter for Filtering Yaw

First, the variance was calculated using the raw yaw data from the file 2020-02-08_08;22;47.csv, where the IMU and GPS were held stationary. The raw yaw data measurements were modeled as normally distributed, as shown in Figure 1 below. From this normally distributed raw yaw data, the variance was determined to be 1.93

Fig.1-Normally Distributed Raw Yaw Data Measurements when the IMU and GPS were held stationary.



Second, a 1D Kalman Filter (KF) was coded in Python to filter raw vehicle yaw angle data and estimate the vehicle's yaw angle by implementing a correction step. This used the normalized raw data variance of 1.93. The equations of Figure 2 and Figure 3 were used to code the Kalman Filter.

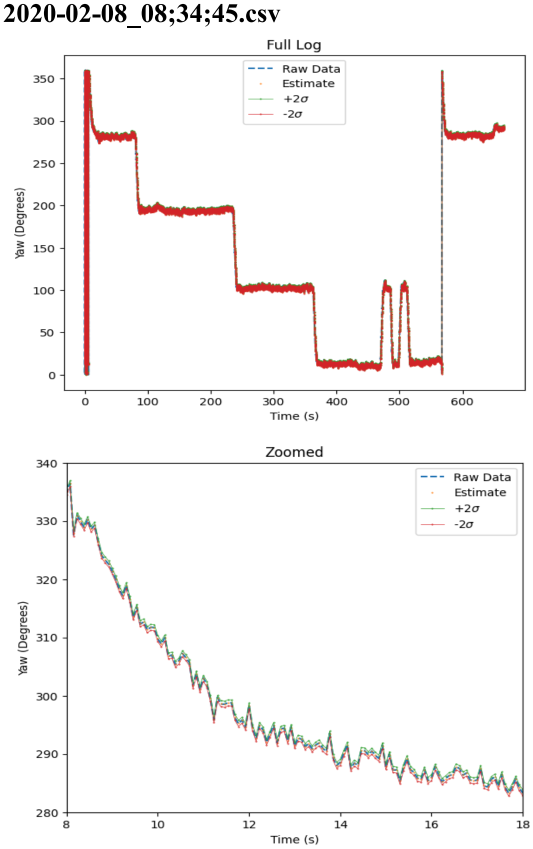
$$\bar{x}_t = a\hat{x}_{t-1} + bu_t$$
$$\bar{\sigma}_t^2 = a^2\hat{\sigma}_{t-1}^2 + b^2\sigma_u^2 \qquad \textbf{[1]}$$

Where state variable estimates $\hat{x}_t$ are normally distributed with variance σ_t^2, control inputs u_t are normally distributed with variance σ_u^2 and the predicted state $\bar{x}_t$ is normally distributed with corresponding variance $\bar{\sigma}_t^2$.

$$K_t = \bar{\sigma}_t^2 / (\bar{\sigma}_t^2 + \sigma_z^2)$$
$$\hat{x}_t = \bar{x}_t + K_t\,(z_t - \bar{x}_t)$$
$$\hat{\sigma}_t^2 = \bar{\sigma}_t^2 - K_t\bar{\sigma}_t^2 \qquad \textbf{[2]}$$

Where Kt is the Kalman Gain, zt is the current measurement, $\hat{x}_t$ is the current state estimate, and $\hat{\sigma}_t^2$ is the corresponding variance. In implementing Eq.1 for the prediction step and Eq. 2 for the correction step, we get the following Figure 2 below, plot the raw data, estimated data, and estimated data +/- 2 σ bounds. The right figure zooms in for only 10 seconds of data to illustrate if the raw and filtered data lie within the 2 bounds. Looking at the zoomed figure on the right of both Fig. 2 and Fig. 3, it appears that both raw and filtered data – which are marked on the legend – do lie within the 2 bounds.

Fig.2 - Raw data and filtered data with +/- 2σ bounds. Full size on top, zoomed in on bottom 2020-02-08_08;34;45.csv



B. *Bayes Filter For Estimating Vehicle is Stopped*

The vehicle data is provided in E205_Lab2_NuScenesData.xlsx, and includes the [x y ϑ] data for seven different vehicles – the ego vehicle and six other vehicles – logged at 0.5-second time steps.

To model the relationship between vehicle speed and stop state, histograms of vehicle speed were created using [x y] data. Vehicle 4, which is parked for the entire duration of the video, was used to construct the empirical conditional probability density function $p(s \mid S)$, where S denotes the stopped state. Vehicles 2, 3 and 5, which are moving, were used to construct the empirical conditional probability density function $p(s \mid N)$, where N denotes the not-stopped state. Gaussian fits were overlaid for parametric approximation, though the histograms better capture skew and outliers.

Fig. 3 shows $p(s \mid S)$ for Vehicle 4. Since Vehicle 4 is stationary, the probability density function peaking near zero meters per second is expected. Fig. 4 shows $p(s \mid N)$ using Vehicles 2, 3 and 5. Although the distribution is left-skewed, the x-axis spacing reveals a peak near approximately 1 m/s, which is consistent with slow vehicle motion.

Fig 3 - p(s|stopped) vs Speed (m/s)

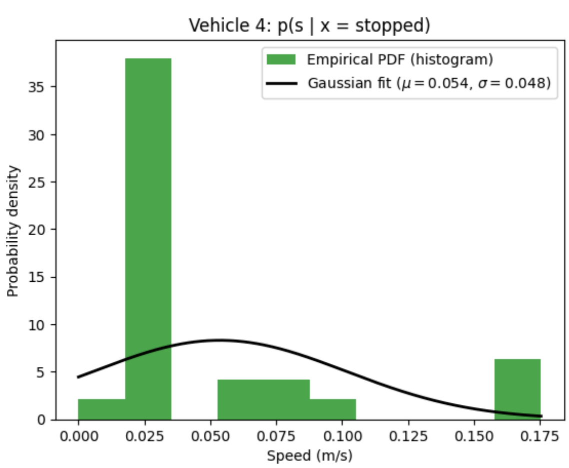
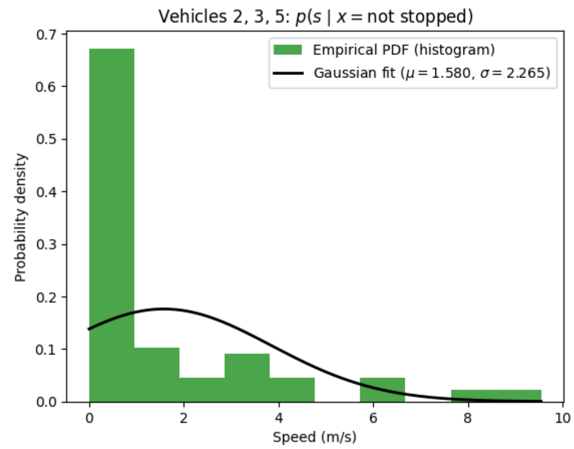


Fig 4 - p(s|not stopped) vs Speed (m/s)



The Bayes Filter is a recursive algorithm consisting of a prediction step and a correction step. It maintains a belief state $bel_t(x_t)$, which represents the probability that the system is in state x_t at time t, given the previous state x_{t-1}. As the vehicle state is binary:

$$x_t \in \{S,\ N\}$$

where S denotes stopped and N denotes not stopped. The state transition probabilities are given by:

$$p\left(x_{i,t} = stopped \middle| x_{i,t-1} = stopped\right) = 0.6$$
$$p\left(x_{i,t} = not\ stopped \middle| x_{i,t-1} = stopped\right) = 0.4$$
$$p\left(x_{i,t} = stopped \middle| x_{i,t-1} = not\ stopped\right) = 0.25$$
$$p\left(x_{i,t} = not\ stopped \middle| x_{i,t-1} = not\ stopped\right) = 0.75 \quad \textbf{[3]}$$

These probabilities encode prior assumptions about how likely a vehicle is to remain stopped, start moving, stop from motion, or continue moving. The prediction step propagates the belief forward in time using the state transition model:

$$\overline{bel}(x_t) = \sum_{x_{t-1}} p(x_t \mid u_t, x_{t-1}) bel(x_{t-1}) \quad \textbf{[4]}$$

For the two-state system, this becomes:

$$\overline{bel}_t(S) = 0.6 bel_{t-1}(S) + 0.25 bel_{t-1}(N) \quad \textbf{[5]}$$
$$\overline{bel}_t(N) = 0.4 bel_{t-1}(S) + 0.75 bel_{t-1}(N) \quad \textbf{[6]}$$

The correction step incorporates the speed measurement s_t using the conditional measurement models $p(s_t \mid S)$ and $p(s_t \mid N)$ obtained for Figs. 3 and 4 :

$$bel(x_t) = \eta p(s_t|x_t)\, \overline{bel}(x_t) \qquad \textbf{[7]}$$

Which corresponds to:

$$bel_t(S) = \eta_t p(s_t \mid S)\, \overline{bel}_t(S) \qquad \textbf{[8]}$$

$$bel_t(N) = \eta_t p(s_t \mid N)\, \overline{bel}_t(N) \qquad \textbf{[9]}$$

Where the normalization factor η_t ensures the beliefs sum to one:

$$\eta = \frac{1}{p(s_t \mid S)\, \overline{bel}_t(S) + p(s_t \mid N)\, \overline{bel}_t(N))} \qquad \textbf{[10]}$$

This recursion produces: $bel_t(S) = p(x_t = S \mid s_{1:t})$ which represents the probability that the vehicle is stopped at time t given all speed measurements up to that time. At each time step, the measurement likelihoods $p(s_t|\text{stopped})$ and $p(s_t||\text{not stopped})$ are obtained by looking up the speed s_t in the corresponding normalized histogram PDF.

Vehicle 4 should have p(stopped) ≈ 1 throughout, as it is stationary. Vehicles 2, 3 and 5 should remain near zero most of the time, as they are in motion.

Fig 5 - Bayes Filter Output: Probability of Being Stopped

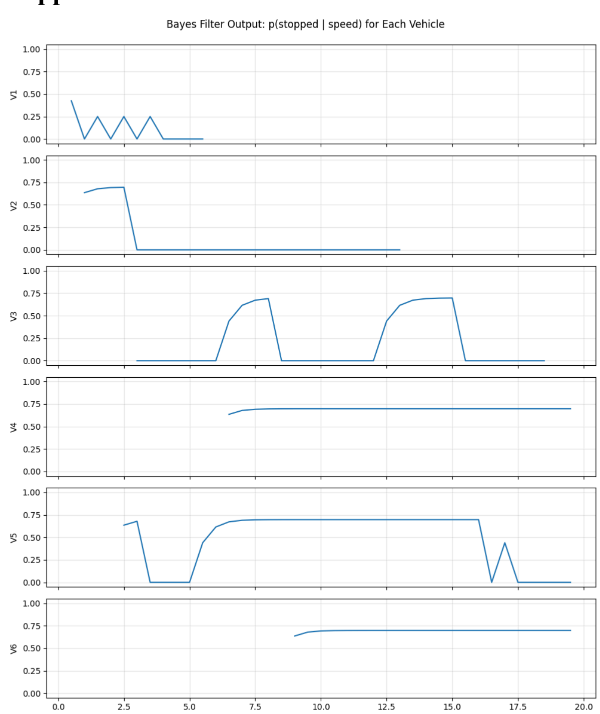
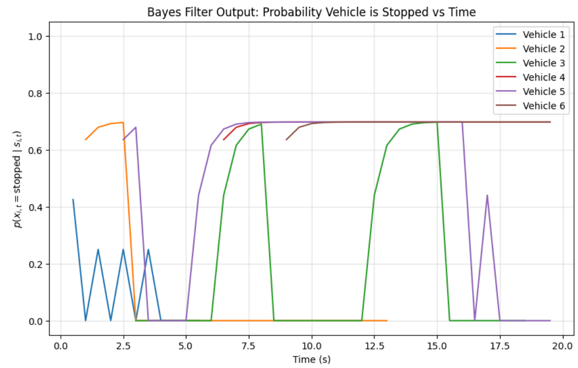


Fig 6 - p(s|not stopped) vs Speed (m/s)



Based on Fig. 5 and the video, Vehicle 6 appears to be parked for the entire duration, but only comes into view around 9 seconds. This vehicle is believed to be the gray car parked on the right side of the street.

C. *Applying Bayes Filter to Police Violence Data*

Using the police violence data from Lab 1, and some US census data, we will compare the American population data to some particular queries of police data using 1 step of a Bayes Filter. According to the US census [3], the probabilities of being certain races in America are shown below in Table 1:

Table 1: Probability of Being a Certain Race in America

White	Asian	Black	Hispanic
0.589	0.063	0.136	0.191

Table 2: Conditional Probabilities p(race | killed by police, armed_status)

Victim Status	White	Black	Hispanic	Asian
Armed	0.5150	0.2788	0.1875	0.0188
Unarmed	0.4511	0.3370	0.1998	0.0121
Unclear	0.4598	0.2804	0.2430	0.0168

A one-step Bayes filter was used to predict the race distribution of police killing victims under two hypothetical armed–status scenarios. The probability rule was applied: $p(race = i \mid pwkbp = true) =$

$\sum_j p(race = i \mid pwkbp = true, armed\_status = j) \cdot$
$p(armed\_status = j)$
where $i \in \{White,\ Black,\ Hispanic,\ Asian\}$ and $j \in \{Armed,\ Unarmed,\ Unclear\}$

The conditional probabilities $p(race = i \mid pwkbp = true,\ armed\_status = j)$ were obtained in Table 5. Two cases were evaluated:

Case (a)
$p(armed\_status = true) = 0.8$
$p(armed\_status = unclear) = 0.2$
Case (b)
$p(armed\_status = false) = 0.8$
$p(armed\_status = unclear) = 0.2$

All other armed-state probabilities were assumed to be zero in each case. Using the law of total probability, the predicted race probabilities were computed for both scenarios and are summarized in Table 3:

Table 3: Predicted Race Probabilities

Race	Case (a)	Case (b)
White	0.5040	0.4528
Black	0.2791	0.3257
Hispanic	0.1986	0.2084
Asian	0.0184	0.0131

These results show how the assumed distribution of armed status influences the predicted race probabilities through the Bayes filter. In Case (a), where most victims are assumed to be armed, the predicted race distribution more closely reflects the empirical race distribution of armed victims from part (b). In Case (b), where most victims are assumed to be unarmed, the predicted race distribution shifts toward the empirical race distribution of unarmed victims.

This demonstrates how the Bayes filter propagates prior assumptions (armed-status probabilities) through conditional likelihoods to produce updated race probability estimates.

Now we implement the Bayes filter correction step by updating the prior race probabilities using the observation that the victim's age is less than 20 years old. The police_killings dataset represents the condition person_was_killed_by_police = true, so all probabilities are implicitly conditioned on pwkbp = true. The likelihood terms were computed empirically from the dataset for each race state i:
$p(age < 20 \mid pwkbp = true,\ race = i)$
These likelihoods represent how frequently age <20 occurs within each race subset of the police_killings dataset. Next, Bayes' Rule was applied to compute the posterior probability of race conditioned on the age <20 observation:
$p(race = i|pwkbp = true,\ age < 20) =$
$\eta p(age < 20|pwkbp = true, race = i)p(race = i|pwkby = true)$

Where $p(race = i|pwkbp = true)$ is the prior distribution obtained from external demographic data in Table 4, and η is a normalization factor ensuring the posterior probabilities sum to one:

$$\eta = \frac{1}{\sum_i p(age<20 \mid pwkbp = true, race = i)p(race = i \mid pwkby = true)} \quad \textbf{[11]}$$

The resulting posterior: $p(race = i|pwkbp = true,\ age < 20)$ provides four updated state estimates (one per race) that quantify how the age <20 observation reweights the prior race probabilities according to the dataset-derived likelihoods.

Table 4: Bayes Correction Step Using Age < 20

Race	p(age<20|race)	Prior p(race)	Posterior p(race|age<20)
White	0.0313	0.6016	0.3852
Black	0.0888	0.1389	0.2524
Hispanic	0.0740	0.1951	0.2954
Asian	0.0508	0.0644	0.0669

References

[1] E205 Lab 2, https://sites.google.com/g.hmc.edu/e205-fa23/labs (accessed Sep. 18, 2023).

[2] E205 Lab 1, https://sites.google.com/g.hmc.edu/e205-fa23/lectures (accessed Sep. 18, 2023).

[3] U.S. Census Bureau quickfacts: United States, https://www.census.gov/quickfacts/fact/table/US/PST045222 (accessed Sep. 18, 2023).

[4] E205 Lab 2 Notebook https://colab.research.google.com/drive/131M0yT_Wk0Vy8JDwTzZy73whIp4U0kkH?usp=sharing (accessed April 12, 2026)